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BIOLOGICAL OBJECT SEGMENTATION ON VIDEO DATA USING OPTICAL FLOW-GUIDED TEMPORAL CONSISTENCY

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ABSTRACT

Biological videos are frequently affected by imaging artifacts, such as noise, motion blur, or low contrast, which can cause individual frames to lack sufficient signal for accurate segmentation. As a result, conventional single-frame approaches frequently miss faint or transient objects. We present FOGNET, a lightweight multi-frame deep learning model that leverages temporal context in order to improve biological object segmentation on video data. Our approach combines simultaneous multi-frame predictions with optical-flow-based warping to enforce temporal consistency, while only requiring sparsely annotated video data for training. FogNet achieves state-of-the-art performance across diverse datasets, demonstrating robustness to noise and large inter-frame displacements. These results highlight the potential of temporal modeling for reliable segmentation in dynamic biological systems.

***Index Terms*—** Biological object segmentation, U-Net, Video data, Optical Flow

1. INTRODUCTION

The first step of most automated biological image analysis pipelines is the segmentation of specific biological objects (cells, pathogens, molecules, ...), allowing their enumeration, tracking[1], or spatial distributions analysis [2]. Objects can vary greatly in size and shape, and range from fluorescent spots to large cells with complex morphologies. Analytical methods specifically tailored to each subclass of objects have therefore been developed, for example, wavelet detection [3] for fluorescent spots, or shape fitting [4] for cell segmentation.

In recent years, deep learning–based segmentation algorithms have introduced more versatile and generalizable solutions. Among them, the U-Net architecture [5] has become a cornerstone in biomedical image analysis, serving as the foundation for numerous extensions. For example, StarDist [6] models star-convex cell shapes, while SplineDist [7] captures more elongated or irregular geometries. Furthermore, these architectures have been adapted for instance segmentation through a variety of strategies, including distance embedding [8], flow-field representations [9] or non-maximum suppression [6].

However, existing deep learning methods typically process video data frame by frame, treating each image as an independent sample rather than as part of a temporally coherent sequence. This limits their ability to detect weak or noise-obscured structures whose presence or shape could be reinforced by information from neighboring frames. Multi-frame object detection methods have been developed in the context of bounding-box–based detections in natural videos, using multi-channel networks [11], motion estimation [12], or temporal attention [13]. However, these methods make use of massive training datasets (Image-Net, CIFAR, ...) available for natural images, require complex architectures, and do not achieve pixel-wise segmentation. We propose a solution to these challenges with a Flow-Guided U-Net (FogNet), a light-weight multi-frame model capable of pixel-level segmentation, all while requiring minimal annotated image data. Our approach is simple to train and requires no data augmentation. The code is publicly available[1]. We benchmark FogNet against leading single-frame segmentation models on a range of biological video datasets and evaluate its robustness using the SINETRA simulator [10] under various noise and motion conditions.

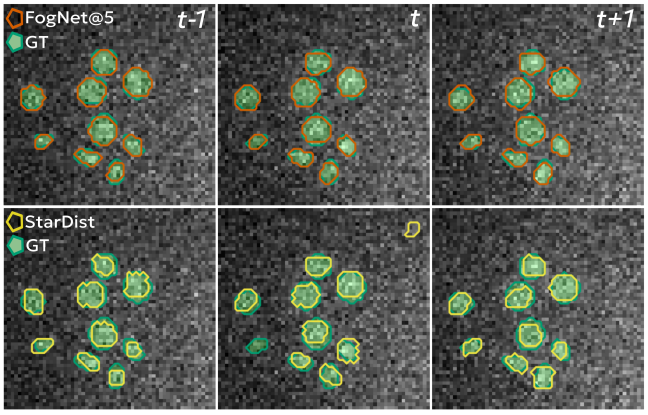


Fig. 1: Example results on zoomed-in regions from 3 consecutive frames of SINETRA's Hydra Flow scenario [10]. Our proposal (top) uses temporal context to produce a smoother and more consistent segmentation across frames compared to the single-frame StarDist method (bottom).

2. METHOD

We propose a flow-guided deep learning framework for temporally consistent segmentation of biological videos. A temporal window centered on each frame is processed by an adapted U-Net producing probability maps for all frames in the window. Optical flow is used to warp these predictions toward the central frame, and a differentiable Gaussian-weighted averaging module combines them into a single temporally aligned output. Final binary and instance masks are obtained through thresholding and standard post-processing.

The next sections describe the architecture and training strategy in detail.

[1] https://github.com/bia-pasteur/FogNet

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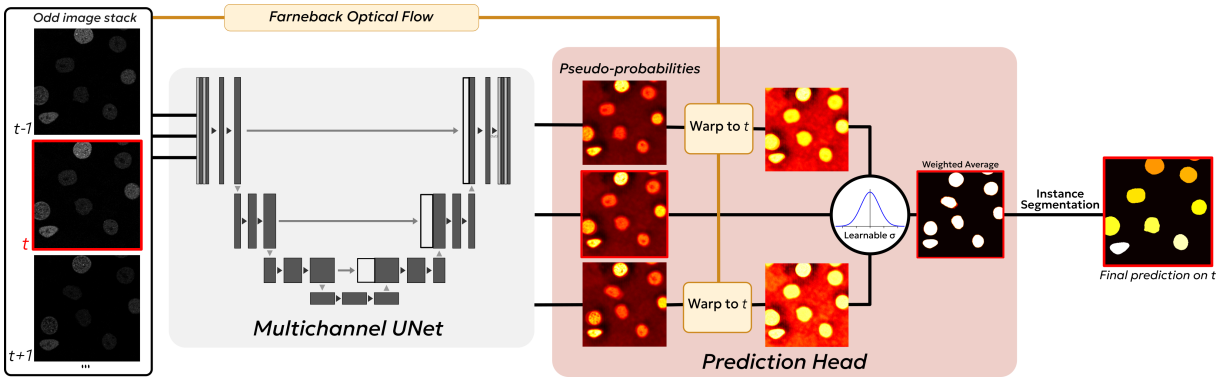


Fig. 2: The Flow-Guided U-Net Architecture. Model input is a odd image stack, and output is the prediction on the central image. The Flow-guided temporal Gaussian averaging is integrated in the prediction head with standard deviation σ as a learnable parameter. Instance segmentation is performed from the predicted probability maps as a post-processing step outside of the deep learning framework.

Problem formulation. Given an input video $V = \{I_t\}$, $t \in [1, \ldots, T]$, our goal is to output a segmentation mask $\hat{y}_t$ for each frame. In our case, let each frame I_t be a single-channel image, as is often the case in biological imaging. We provide the model with a odd-numbered stack of frames, which we call a temporal window: $W_t^{2n+1} = [I_{t-n}, ..., I_t, ..., I_{t+n}]$. For training, our model only requires the annotation of the central image y_t. Edges cases ($t \leq n$ and $t > T - n$) are dealt with by padding the video beginning and end with n instances of the edge frames: $V = [I_1, ..., I_1, ..., I_T, ..., I_T]$.

2.1. Model architecture

The U-Net backbone. We adapt the U-Net architecture to accept the temporal window as a single input with $2n + 1$ channels. The final convolution layer is modified to output $2n + 1$ channels, corresponding to the predicted probability map $P_{t,i}, i \in [t - n, t + n]$ for each frame of the temporal window. Each map $P_{t,i}$ represents a pseudo-probability distribution over pixels, indicating the model's confidence that a given pixel belongs to an object rather than the background.

Optical Flow Warping. To align pseudo-probabilities across frames, we estimate optical flow within the temporal window. Each flow map $F_{t\to t+1}$ represents pixel displacements between I_t and I_{t+1}, allowing I_t to be reconstructed by warping I_{t+1} using $F_{t\to t+1}$. For a window W_t^{2n+1}, the set of flows is

$$[F_{t-n+1\to t-n}, \ldots, F_{t\to t-1}, F_{t\to t+1}, \ldots, F_{t+n-1\to t+n}].$$

These flows warp the probability maps predicted by the U-Net head. Each prediction $P_{t,i}$ is iteratively warped to the central frame, producing the aligned map $\hat{P}_{t,i}$, as detailed in Algorithm 1.

The optical flow in FogNet is estimated using the Farnebäck algorithm [14], which offers a good balance between computational efficiency and accuracy for dense motion estimation. Nevertheless, the optical flow module in FogNet is fully modular and can be replaced by alternative methods depending on the application's precision and performance requirements.

Prediction Weighted Averaging Head. The final trainable module is a prediction head that fuses the warped predictions using Gaussian-weighted averaging based on their temporal distance from the central frame. This approach emphasizes the central prediction while smoothly modulating the influence of neighboring ones through a learnable Gaussian weighting function $\mathcal{N}(i; t, \sigma)$. Each prediction receives a weight w_i according to its position in the temporal window (Equation 1):

$$\hat{P}_t = \frac{1}{\sum_{i=t-n}^{t+n} w_i} \sum_{i=t-n}^{t+n} \hat{P}_{t,i} w_i, \quad w_i = \mathcal{N}(i; t, \sigma) \qquad (1)$$

This operation is fully differentiable, enabling the standard deviation σ to be learned during training. The resulting map $\hat{P}_t$ constitutes the final flow-guided prediction for the central frame.

Instance Segmentation. We threshold $\hat{P}_t$ to obtain a binary mask and perform instance segmentation to distinguish individual objects. For isolated objects, connected component analysis suffices. For touching objects, we instead apply watershed segmentation using the maxima of the probability map as seeds to separate adjacent instances, enabling accurate counting, and shape analysis beyond semantic segmentation.

Algorithm 1 Flow Warping for Probability Maps

Input $[P_{t,t-n}, ..., P_{t,t+n}]$ ▷ U-Net output for W_t^{2n+1}
for $i \in [t - n, t + n]$ **do**
  $\hat{P}_{t,i} \leftarrow P_{t,i}$
  if $i < t$ **then**
    while $i < t$ **do**
      $\hat{P}_{t,i} \leftarrow \mathcal{W}(\hat{P}_{t,i}, F_{i+1\to i})$ ▷ Warp to center
      $i \leftarrow i + 1$
    end while
  end if
  if $i > t$ **then**
    while $i > t$ **do**
      $\hat{P}_{t,i} \leftarrow \mathcal{W}(\hat{P}_{t,i}, F_{i-1\to i})$ ▷ Warp to center
      $i \leftarrow i - 1$
    end while
  end if
  return $\hat{P}_{t,i}$ ▷ $\hat{P}_{t,i}$, a reconstruction of P_t
end for
Output $[\hat{P}_{t,t-n}, ..., \hat{P}_{t,t+n}]$

2.2. Model Training

For fully annotated data, the model is trained on 200 frames from a single video over 6000 steps with batch size 1. The temporal window

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size is a hyperparameter. U-Net weights are initialized randomly, and the Gaussian aggregation standard deviation $\sigma \in]0, \sigma_{max}[$ is learned via $\sigma_{param} \in]-\infty, +\infty[$, initialized so that $\sigma = n$:

$$\sigma = \mathrm{Sig}(\sigma_{param}) \times \sigma_{max},$$

with

$$\mathrm{Sig}(x) = \frac{1}{1+e^{-x}}, \text{ and } \quad \sigma_{max} = \frac{n}{\sqrt{-2\log(1-10^{-2})}}.$$

Two Adam optimizers are used: one for the U-Net backbone, following a standard warm-up and decay schedule, and another for the σ parameter, whose learning rate progressively increases during training (Fig. 3). This scheduling allows the model to first stabilize pixel-level predictions before learning temporal aggregation. For sparsely annotated videos, only the central frame of each temporal window needs labels, allowing the model to leverage neighboring unlabeled frames. Validation is performed every 200 steps for early stopping and threshold optimization.

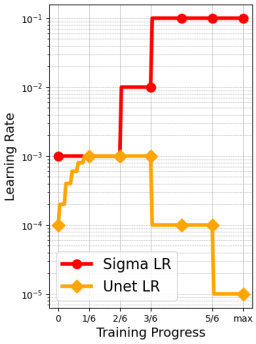


Fig. 3: Learning rate schedulers.

Training was performed on an NVIDIA RTX 3500 Ada GPU (12 GB VRAM); with temporal window of 5 frames on a $[200, 1024, 1024]$ video, training takes 80 min, and predictions run at 1 fps.

3. EXPERIMENTS

3.1. Datasets

We evaluate our approach on three distinct datasets that cover complementary aspects of our problem, ranging from fully synthetic data with perfect annotations to real biological recordings with sparse or dense ground truth. This diversity allows us to assess both the robustness and generalization capabilities of our method.

SINETRA: Synthetic 2D+T videos of fluorescent particles on a deformable background [10]. We can either simulate the movement of *Hydra vulgaris* (Hydra Flow scenario) or model random elastic deformations (2D Springs scenario). The Signal-to-Noise Ratio of the video can be adjusted. Videos are fully annotated, with shapes of $[200, 1024, 1024]$. We generate three videos for each scenario: one for training, one for validation, and one for testing.

Hydra vulgaris: Real fluorescence time-lapse videos of behaving *Hydra vulgaris* [15]. Targets correspond to neuronal nuclei. The data is extremely sparsely annotated, with only 12 independent $[200, 200]$ annotated patches extracted from three $[1000, 1024, 1024]$ videos. For this dataset, we train on 8 annotated patches, use two for validation, and two for testing, repeating the experiment across different splits.

GFP-GOWT1: Fluorescence videos of behaving mouse stem cells [16]. The dataset is fully annotated and consists of two videos with shapes $[92, 1024, 1024]$. One is used for training and the other for testing, without a separate validation set.

3.2. Compared Methods

U-Net: We train a standard two-class U-Net architecture [11] on all datasets. Instance segmentation is obtained through a dedicated post-processing step, following the same procedure as for FogNet.

StarDist: This architecture extends U-Net by predicting star-convex polygons to delineate individual cells [6]. Previous work [15] has shown that StarDist achieves state-of-the-art performance on *Hydra vulgaris* videos and is particularly effective on the GOWT1 dataset, where cell morphologies are approximately star-convex. We therefore include this widely adopted architecture as a strong baseline for comparison, retraining it using its default training settings provided in the official implementation.

Cellpose: Based on a gradient-flow representation of object boundaries, Cellpose [9] is a deep learning model for cell segmentation known for its strong generalization capabilities. For the SINETRA and *Hydra vulgaris* datasets, we retrain Cellpose from scratch using its default configuration,, while for GOWT1 we fine-tune the pretrained Cellpose-SAM model to better adapt to our data.

FogNet@k: Proposed model with a temporal window $k = 2n + 1$.

3.3. Evaluation Metric

We compute the average precision $AP_\tau = \frac{TP_\tau}{TP_\tau + FN_\tau + FP_\tau}$ for different Intersection-over-Union (IoU) thresholds τ, following [6, 9]. The final AP score is obtained by averaging the results over five experiments with different random initialization seeds.

In addition, we measure the binary (object-versus-background) IoU to assess the overall quality of the segmentation, independently of the instance separation step. This metric isolates the pixel-level accuracy of object delineation from the performance of the instance identification procedure described in Section 2.1.

4. RESULTS

Qualitative Analysis. We begin by visually assessing model outputs across both simulated and experimental datasets. As shown in Fig. 5, FogNet produces fewer false positives, sharper boundaries, and overall higher segmentation quality compared to baseline methods. Furthermore, Fig. 1 demonstrates that FogNet preserves temporal consistency across consecutive frames, maintaining coherent object detections over time. These observations confirm that explicit temporal modeling enhances both detection reliability and segmentation stability.

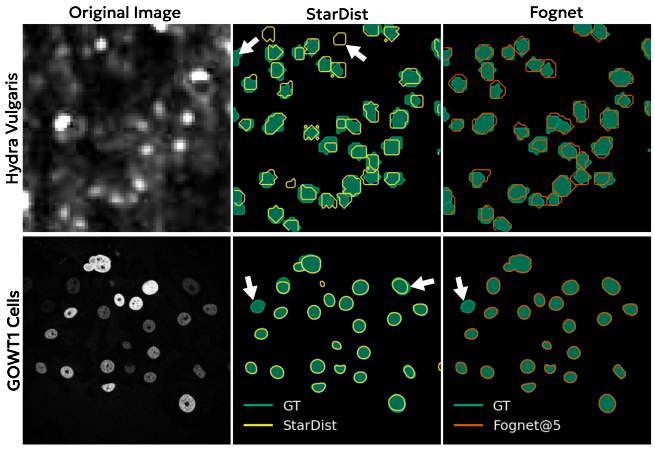


Fig. 5: Example detections with StarDist and FogNet on zoomed-in frames of *Hydra vulgaris* (top) and GOWT1 cells (bottom). Ground-truth objects are shown in green. White arrows highlight common segmentation errors.

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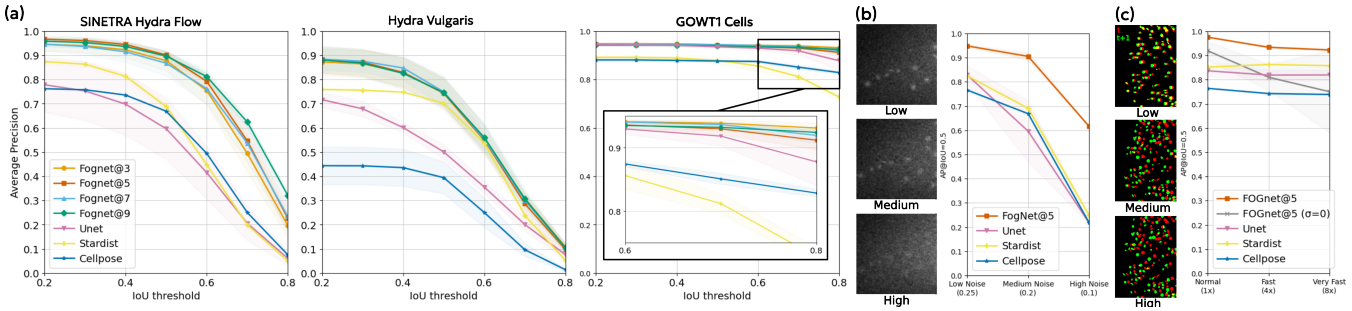


Fig. 4: **(a)** Comparisons of detection scores between FogNet with different temporal window sizes and baseline models on SINETRA's Hydra Flow scenario, real *Hydra vulgaris* annotated patches and GOWT1 cells videos (left to right). **(b)** Comparisons of APs for an IoU threshold of 0.5 between FogNet with a temporal window of 5 frames and baseline models for different Signal to Noise ratios on SINETRA's Hydra Flow scenario. **(c)** Comparisons of APs for an IoU threshold of 0.5 between FogNet with a temporal window of 5 frames, its ablated version with $\sigma = 0$, and baseline models for different movement speeds on SINETRA's Springs 2D scenario.

Model	Hydra Flow	Hydra Flow High Noise	*Hydra vulgaris*	GOWT1 cells
U-Net	62.2% ± 4.7%	36.5% ± 0.4%	60.5% ± 1.4%	90.6% ± 0.9%
StarDist	62.5% ± 1.0%	34.5% ± 0.1%	60.6% ± 2.1%	85.7% ± 0.5%
Cellpose	59.5% ± 0.6%	28.2% ± 0.5%	38.7% ± 6.2%	**91.5% ± 0.2%**
FogNet@5 (Ours)	**76.9% ± 2.1%**	**61.7% ± 0.5%**	**67.9% ± 3.4%**	**91.6% ± 0.6%**

Table 1: Binary IoU Scores of compared models on different datasets

Detection Performance. On the Hydra Flow scenario of the SINETRA simulator, our model significantly outperforms both StarDist and U-Net (Fig. 4(a)), achieving near-perfect average precision at low IoU thresholds. Since target particles occupy only a few pixels, performance drops sharply at higher IoU thresholds for all models, as even single-pixel deviations induce large relative errors. When applied to experimental datasets, FogNet again outperformed competing methods in detection accuracy. On the GOWT1 dataset, FogNet maintained strong AP even at high IoU thresholds, reflecting its robustness in localizing cell boundaries with high spatial precision. Increasing the temporal window slightly improves accuracy, but we adopt FogNet@5 for most experiments as it provides an optimal balance between detection accuracy and computational cost.

To evaluate robustness under different noise conditions, we varied the SINETRA noise parameter α, which controls the signal-to-noise ratio (SNR). Three conditions were tested—low, medium ($\alpha = 0.2$), and high noise—and FogNet consistently outperformed baseline architectures across all levels, with the largest performance gain observed under high noise (Fig. 4(b)). These results demonstrate that FogNet maintains reliable detection even in challenging, low-SNR environments.

We next assessed the model's ability to leverage temporal context under fast motion using the 2D Springs scenario, which simulates elastic deformations. By down-sampling frames, we generated three motion regimes: normal (every frame), fast (every 4 frames), and very fast (every 8 frames) (Fig. 4(c)). FogNet consistently achieved higher detection accuracy than competing models, even when inter-frame displacements were large.

To further isolate the contribution of temporal aggregation, we evaluated an ablated version of FogNet in which the Gaussian weighting parameter σ was fixed to zero during training. In this configuration, the model relies solely on the central-frame prediction, without fusing information across time. As shown in Fig. 4(c), this ablated model exhibits reduced performance and higher variability between runs, particularly under large displacements. These results highlight the importance of adaptive temporal weighting, showing that aggregating information across neighboring frames is essential for maintaining stable and accurate detections under large inter-frame movements.

Segmentation Quality. To evaluate segmentation independently of instance matching, we computed the mean Intersection-over-Union (IoU) on binary foreground–background masks (Table 1). Despite its lightweight design, FogNet achieves state-of-the-art segmentation performance on both simulated (Hydra Flow) and real *Hydra vulgaris* videos, outperforming StarDist, U-Net, and Cellpose by a substantial margin. On experimental datasets, our model also generalizes well across distinct biological morphologies, confirming its adaptability.

5. CONCLUSION

In this paper, we introduced *FogNet*, a deep learning architecture that integrates multi-frame detection with optical-flow-based warping to achieve state-of-the-art performance in biological object segmentation from video data. Our approach demonstrates strong robustness to both high noise levels and large inter-frame displacements. In contrast to methods that require fully annotated video sequences [12], FogNet can be trained using annotations from only a few selected frames, making the annotation process substantially more practical for experimental users.

In future work, we plan to extend this framework by incorporating a learnable instance segmentation module and by exploring more advanced motion estimation networks [17]. Furthermore, the proposed multi-frame strategy could be particularly valuable for tracking applications, where the additional temporal consistency provided by FogNet has the potential to significantly enhance long-term tracking accuracy and stability.

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None of the authors declare to have a financial conflict of interest in the results of this study.

7. COMPLIANCE WITH ETHICAL STANDARDS

This is a numerical simulation study for which no ethical approval was required.

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